

TWO BOOKS • FDR SUMMARY — capstone lecture deck

One Benjamini–Hochberg correction across every domain

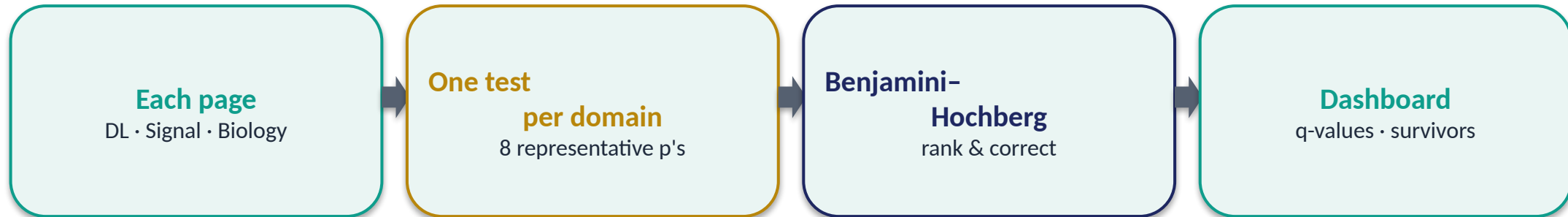
The honest spine

Collect one representative permutation test from each Two Books domain and correct the whole battery together. After correction the cross-domain structure is ROBUST — 6 of 8 tests survive 5% FDR — while the borderline shared-theme claim correctly drops out. Crucial caveat: FDR controls for MULTIPLICITY, not for the sūra-length confound.

Why this is the capstone

Disjoint Letters, Signal and Biology each run many tests. Read alone, the best p from a large battery is how chance masquerades as discovery. This deck reads them ALL together — every q computed live from Book6.

How the dashboard is built



One pipeline, one correction

Each Two Books page contributes one representative permutation test; the shared stats kernel assembles the eight p-values; a single Benjamini-Hochberg pass corrects them together; the dashboard reports q-values and survivors. No page logic is duplicated.

The problem — run many tests, some pass by luck

At $\alpha = 0.05$

1 in 20 NULL tests 'passes' by chance alone.

Run a battery

With 8 tests, the chance of ≥ 1 false hit is far above 5%.

Why an uncorrected 0.05 is unsafe across a battery

Every Two Books page runs several permutation tests. Cherry-pick the smallest p and you will almost always find one under 0.05 — even if nothing real is there. That is the engine of pseudo-miracles.

The shared-theme test lands at $p = 0.049$ — the textbook borderline result multiplicity should make us distrust until it is corrected.

The fix — Benjamini-Hochberg controls the false-discovery rate

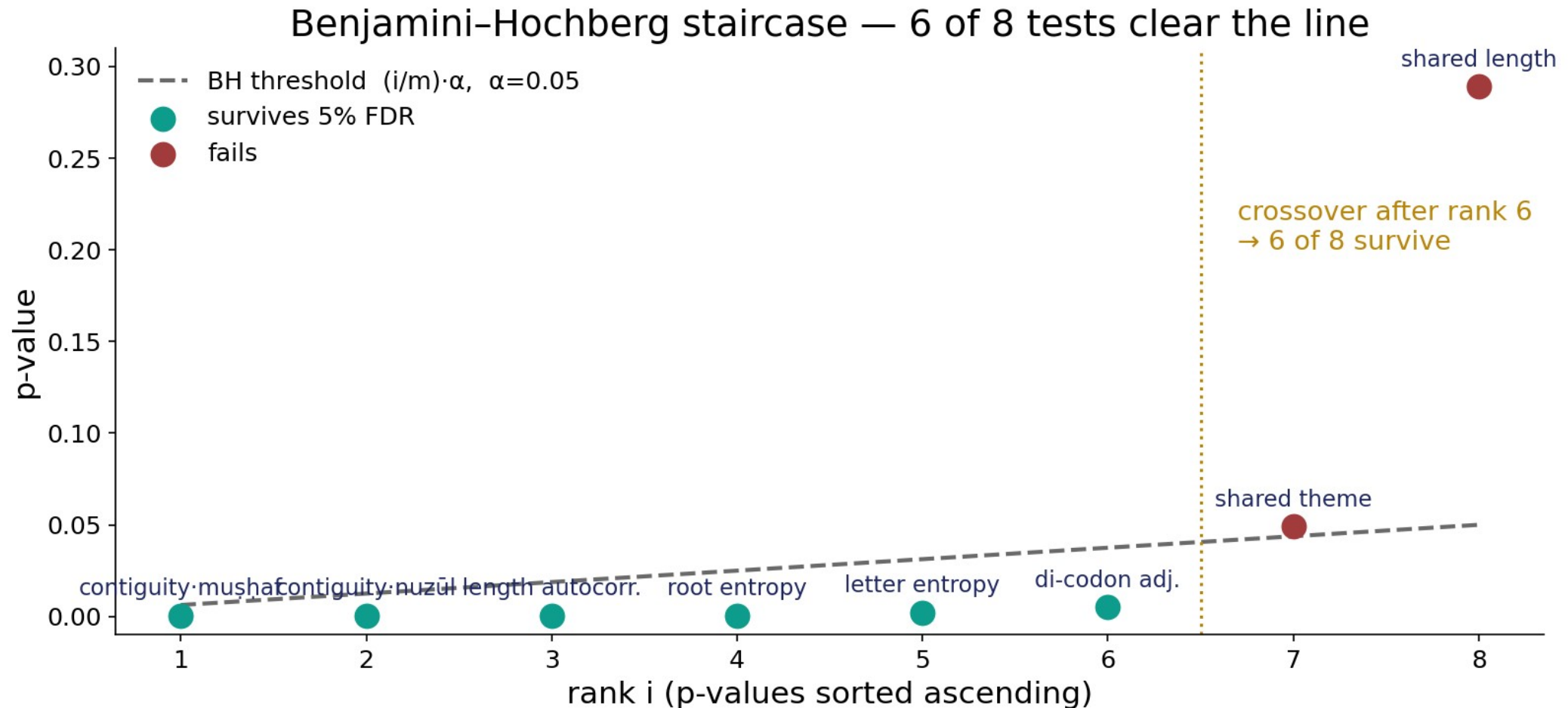


How it works

Rank the m p -values ascending. The i -th passes if it is below $(i \div m) \times \alpha$. The q -value is the smallest α at which a test still passes — a number directly comparable to the 5% threshold.

FDR is the right control for an exploratory battery: less brutal than Bonferroni, yet it still guards against multiplicity. With 8 tests at $\alpha = 0.05$, a raw p of 0.0005 becomes $q = 0.0010$, while $p = 0.049$ becomes $q = 0.056$ — now just over the line.

Module 3 — The ranking, made concrete



In the data — sorted p-values vs the $(i/m) \cdot \alpha$ line. Ranks 1–6 sit below their thresholds and pass; rank 7 (theme, $p = 0.049$) crosses above and fails. Six of eight survive — exactly the q-values.

What this deck will — and will not — do

WILL

- Assemble one representative test per domain.
- Apply ONE BH-FDR across all of them.
- Read each q against 5%.
- Report survivors AND casualties.
- State the method's limit.

WILL NOT

- Read any single p in isolation.
- Promote a borderline uncorrected result.
- Claim FDR fixes the length confound.
- Treat a surviving test as proof of intent.

Module 1 — Frame: 8 tests across 5 domains

What it is / Why

The capstone gathers one representative test from each Two Books domain — eight in all — so a single correction can speak for the whole project, not one page.

Position

Sequence

Semantic

Signal

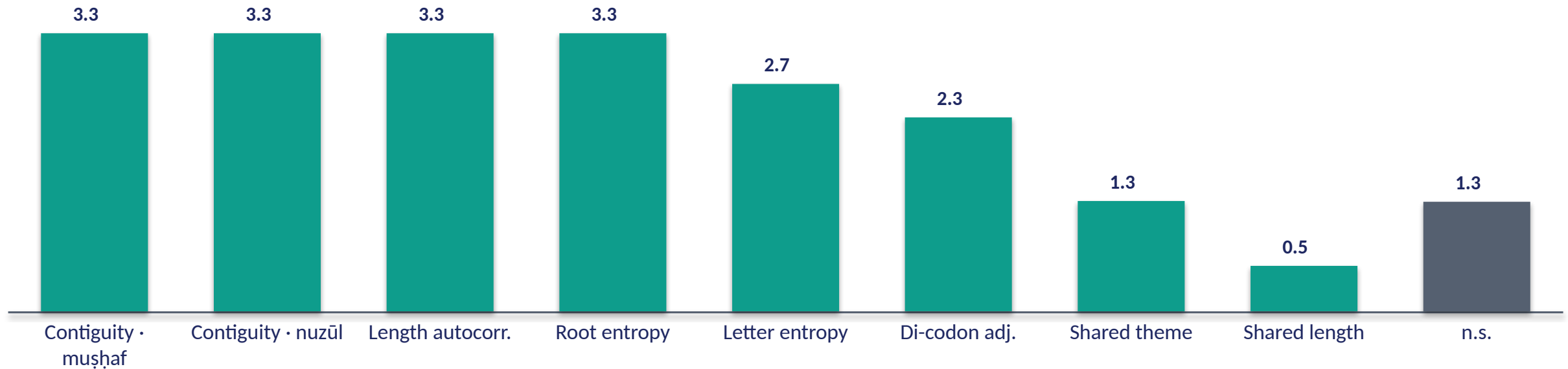
Biology

In the data • Bridge

Position contributes three (contiguity $\times 2$, shared length), Semantic two (theme, root-entropy), and Sequence, Signal, Biology one each. Next: the raw p's themselves.

Module 4 — The battery: one test per domain

Raw permutation p ($-\log_{10} p$) — every domain represented; n.s. line at 1.3

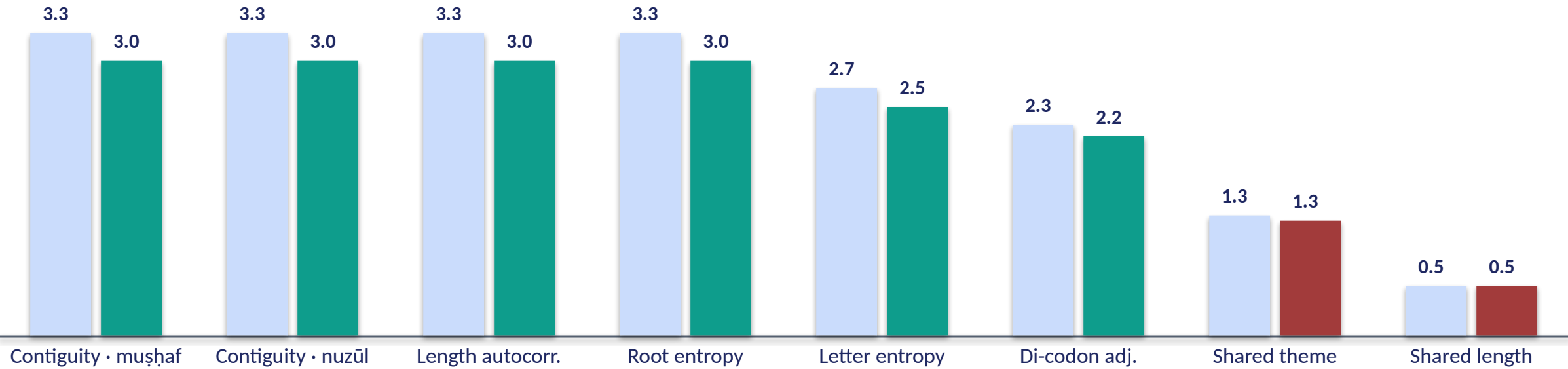


Eight representative tests

Contiguity in two orderings (Position), shared length & shared theme per tag, letter- and root-entropy specials (Sequence/Semantic), length autocorrelation (Signal), and di-codon adjacency (Biology) — one well-chosen test per domain, so the correction never double-counts.

Modules 5-6 — Raw p vs BH-FDR q across all domains

$-\log_{10}$: light bar = raw p, coloured = BH-FDR q (green survives, red fails); 5% line ≈ 1.3

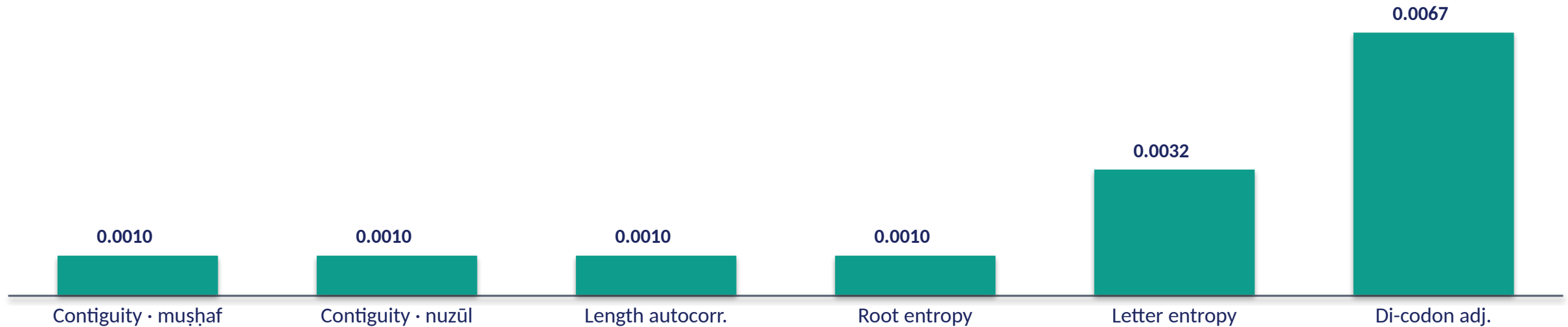


The whole verdict in one picture

Green q-bars clear the 5% line; red q-bars (shared theme, shared length) fall below it. Six of eight survive after correction — the structure is robust; the borderline theme claim is not.

Module 5 — The survivors (6 of 8)

Survivors — BH-FDR q-value (all ≤ 0.0067)

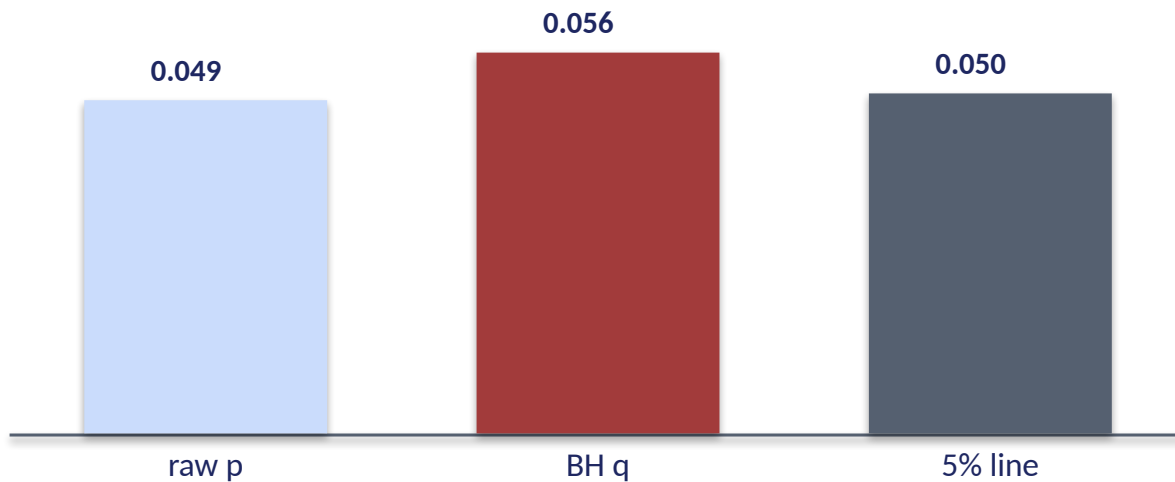


In the data — one survivor from every domain

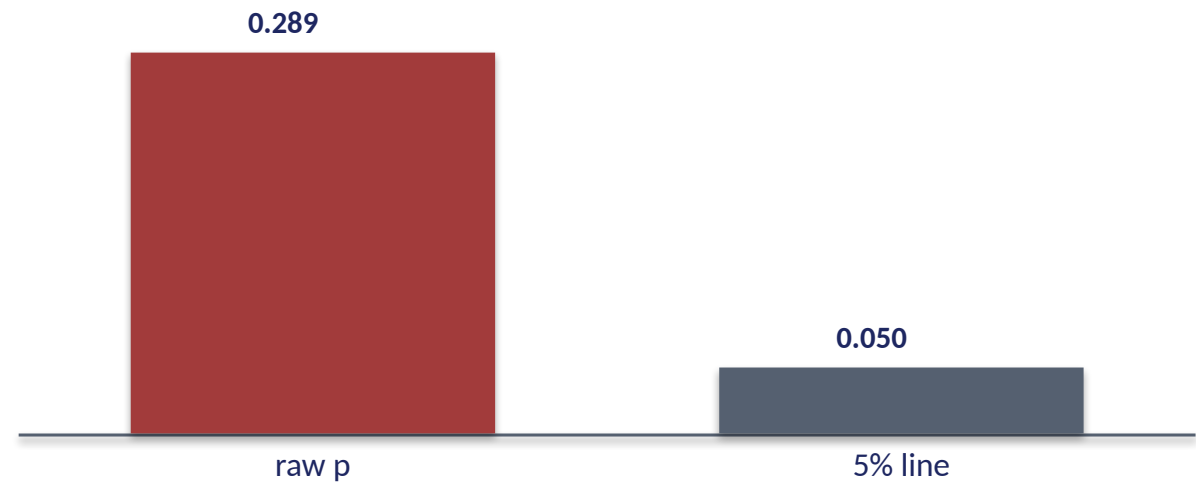
Contiguity in BOTH orderings (q 0.0010 each), length autocorrelation (q 0.0010, Signal), root-entropy (q 0.0010, Semantic), letter-entropy (q 0.0032, Sequence), and di-codon adjacency (q 0.0067, Biology). Every Two Books domain is represented among the survivors.

Module 6 — The casualties: what drops out

Shared theme — raw p vs BH q



Shared length — raw p



The correction works

Shared theme is the textbook case: $p = 0.049$ looks significant alone, but $q = 0.056$ — just over the line. Multiplicity correctly demotes it.

Never close

Shared length per tag is $p = 0.289$ — well inside the null. Both casualties drop out, exactly as a working guard should.

Why $p = 0.049$ is not a discovery

the same number, read alone vs read in a battery

Read ALONE

$p = 0.049 < 0.05 \rightarrow$ looks like a finding.

Read in the BATTERY

$q = 0.056 > 0.05 \rightarrow$ NOT a discovery.

Multiplicity changes the honest reading

A lone borderline p ignores the other seven tests. Benjamini-Hochberg asks how surprising this p is GIVEN the whole battery — and at rank 7 of 8 it is no longer surprising enough. The shared-theme claim is reported as not-supported.

Module 7 — Cross-domain reading

Position

contiguity ×2
(two orderings)

Sequence

letter
entropy

Semantic

root
entropy

Signal+Bio

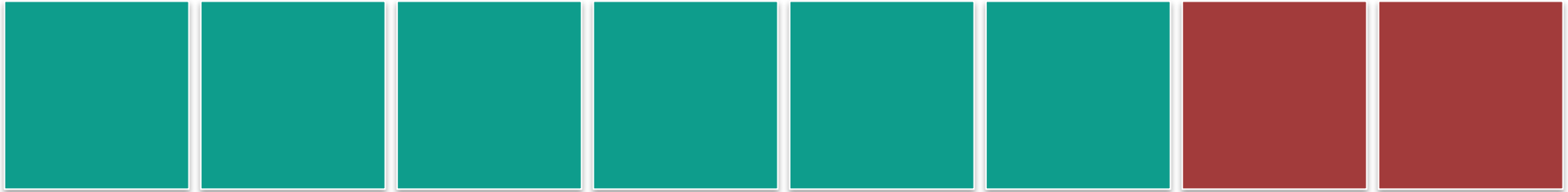
autocorr +
di-codon

Structure that survives independent readings

The six survivors span two **INDEPENDENT** orderings (muṣḥaf and nuzūl contiguity) and all three pages plus the Signal and Biology structure tests. Structure that holds across independent readings and domains is far harder to explain away than any single result.

The corrected dashboard at a glance

green = survives 5% FDR • red = drops out



Six green, two red

Contiguity ×2 • length autocorrelation • root-entropy • letter-entropy • di-codon — survive. Shared theme • shared length — drop out. One row, the whole Two Books verdict.

Takeaway — robust, multi-domain structure under one joint correction; the borderline claim correctly excluded. Bridge: the synthesis names the method's limit.

Synthesis — one corrected dashboard

Survives 5% FDR (6/8)

Contiguity ×2, length autocorrelation, root-entropy, letter-entropy, di-codon — $q \leq 0.0067$, one from every domain.

Drops out

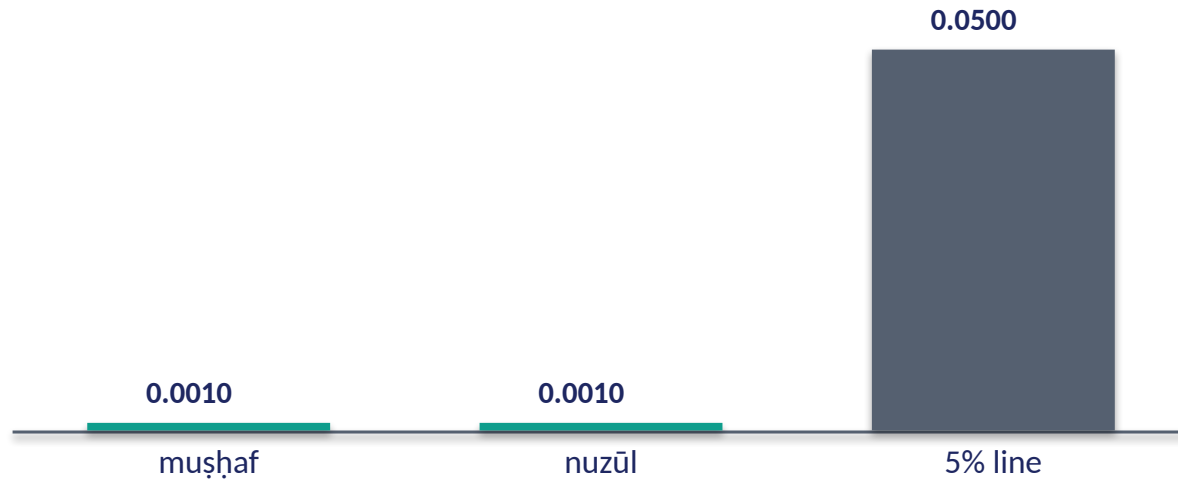
Shared theme ($p\ 0.049 \rightarrow q\ 0.056$) and shared length ($p\ 0.289$). The correction does its job.

The limit

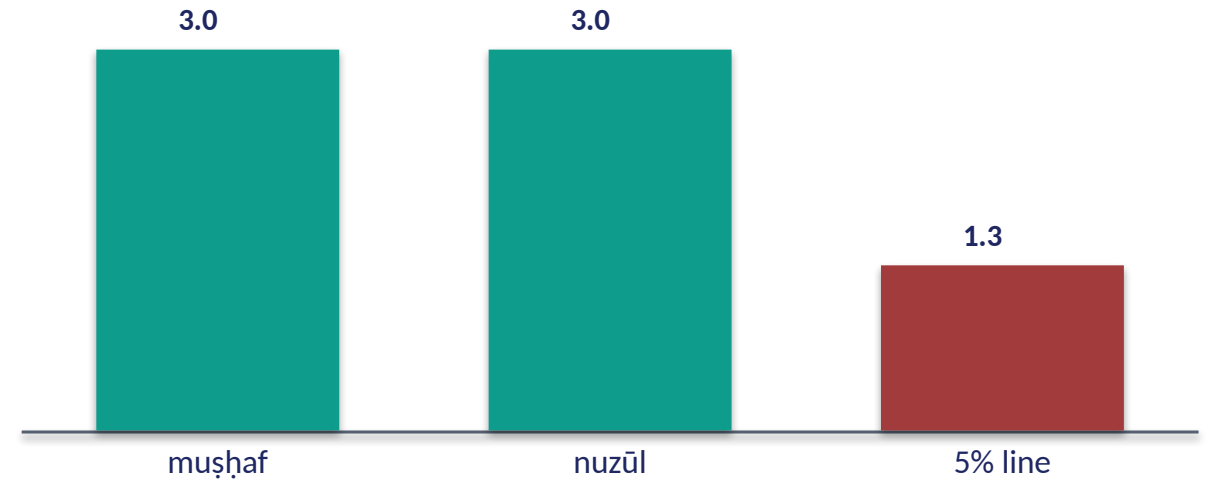
FDR controls MULTIPLICITY — not the sūra-length confound. A surviving test is reproducible, never a proof of intent.

Robustness — the contiguity signal in two orderings

Contiguity q-value — by ordering



Significance ($-\log_{10} q$)



Why two orderings matter

The same contiguity test is run in the muṣḥaf order and the revelation (nuzūl) order — two INDEPENDENT arrangements of the same sūras.

In the data

Both survive at $q = 0.0010$. A signal that holds in two independent orderings is the hardest kind to dismiss as an artefact — the backbone of the survivor set.

Audit — the method, claim by claim

✓ Multiplicity controlled

One BH-FDR across all 8 tests.

✓ Cross-domain robustness

6/8 survive, one per domain.

✓ Borderline demoted

Theme p 0.049 $\rightarrow$ q 0.056, excluded.

✓ Reproducible

Every q re-runs live on the page.

✗ FDR $\leftrightarrow$ confound control

FDR does NOT fix the length confound.

~ Survives $\leftrightarrow$ proof

Reproducible $\neq$ proof of intent.

Not a scientific miracle — and not evidence

What we are NOT claiming

Surviving 5% FDR does not prove design, intent, or a miracle. FDR controls only for **MULTIPLICITY** — it does not correct for the sūra-length confound that several tests share, and it cannot turn a reproducible pattern into evidence of purpose.

What the dashboard IS for

A single, honest place to read the whole Two Books battery together, with multiplicity controlled and every q reproducible live from Book6. It tells us which structure is robust across domains — and which borderline claim to set aside. Judged by reproducibility, never offered as proof.

Quick reference — terms & live numbers

Terms

Multiplicity — many tests inflate false positives. False-discovery rate — expected fraction of 'discoveries' that are null. Benjamini-Hochberg — rank p's, compare to $(i/m) \cdot \alpha$. q-value — smallest α at which a test still passes.

Live Book6 numbers

8 tests · 6 survive 5% FDR · survivors $q \leq 0.0067$ · contiguity $p 0.0005 \rightarrow q 0.0010$ · theme $p 0.049 \rightarrow q 0.056$ (fails) · shared length $p 0.289$ (fails).

Close — the Two Books series, corrected

The through-line

Across Disjoint Letters, Signal and Biology, the corpus shows real geometric, sequential and compositional structure. Read together under one Benjamini–Hochberg correction, six of eight representative tests survive 5% FDR — one from every domain — while the borderline shared-theme claim correctly drops out.

That is the disciplined verdict of the whole series: robust, multi-domain, reproducible structure — with multiplicity controlled, the length confound flagged, and nothing offered as proof of intent.

Every q-value on these slides re-runs live on the [? Two Books · Global FDR](#) page — anyone can reproduce the dashboard from Book6.